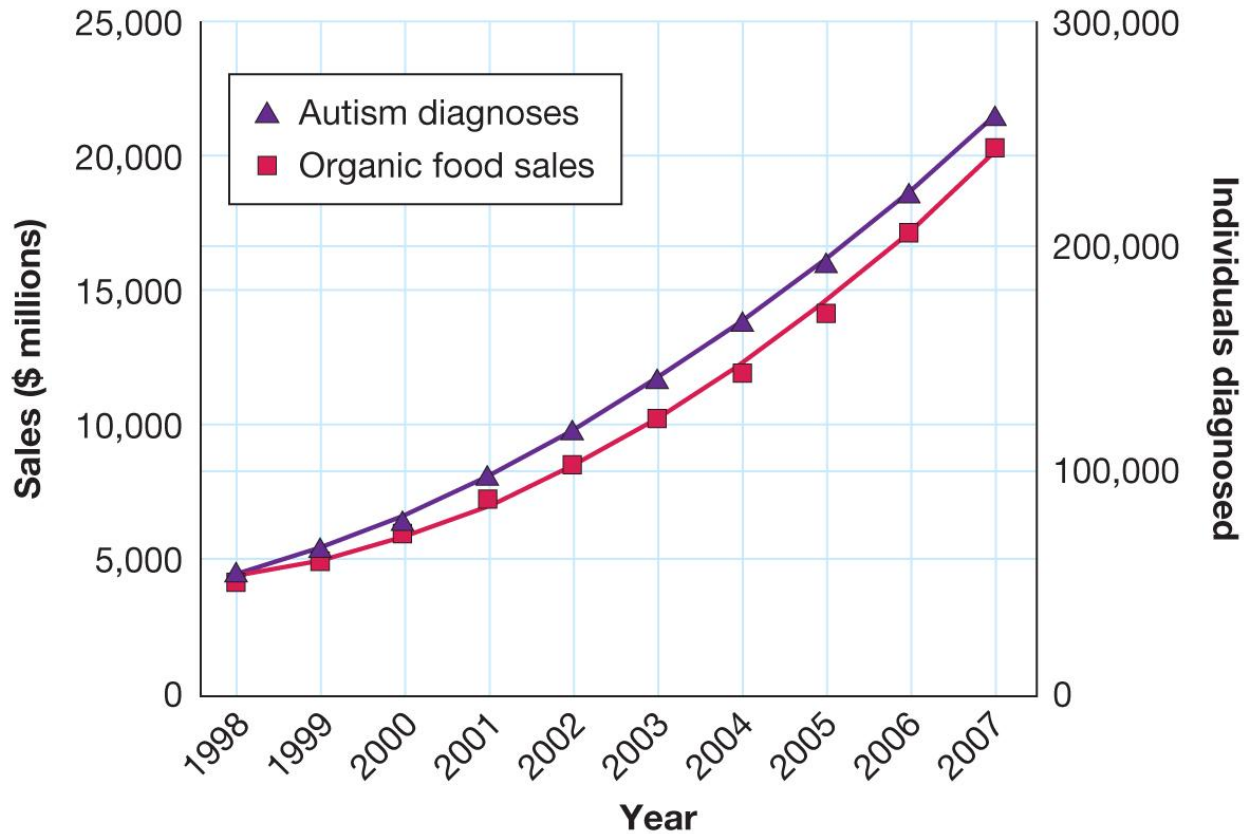


Assignment 9: Bad Data Visualization Analysis

2025-12-15

1. Bad Data Visualization Example For this assignment, I selected a famous example of “Spurious Correlations” from the real world. This graph was originally created to show how data can be manipulated to imply false relationships. • Source: The dataset and graph are from Tyler Vigen’s “Spurious Correlations” project. • Link: <https://www.tylervigen.com/spurious-correlations> • Description: The line chart displays data from the years 1998 to 2007. It compares two completely unrelated variables:
2. Organic food sales (in millions of dollars) - shown with the red line.
3. Number of individuals diagnosed with autism - shown with the purple line.



2. The Problems in This Plot There are several major statistical and visual problems in this chart: • Spurious Correlation (Sahte Korelasyon): The most critical problem is that the graph suggests a cause-and-effect relationship between eating organic food and autism. The graph implies that because organic food sales increased, autism diagnoses also increased. However, in statistics, we know that “correlation does not imply causation.” These two trends are increasing due to other factors (like population growth or better diagnosis methods), not because they affect each other. • Misleading Dual Y-Axes (Yaniltıcı Çift Y Eksen): The graph uses a dual y-axis layout. o The left axis shows sales (0 to 25,000 million \$). o The right axis shows the number of people (0 to 300,000). The scales were manipulated intentionally to make the two lines overlap perfectly. If the axes were scaled differently, the correlation would not look this strong visually.

3. Discussion: How to Solve These Problems To fix this bad visualization and present the data honestly, we can apply the following solutions:
- Separate the Graphs: Since there is no logical link between these two variables, they should not be on the same chart. The best solution is to show them in two separate bar charts or line charts.
 - Avoid Dual Axes for Unrelated Data: We should avoid using dual y-axes for variables that have different units (money vs. people) because it confuses the reader and forces a visual connection that does not exist.
 - Statistical Context: As a data analyst, instead of just looking at the lines, we should check the statistical significance. A scatter plot could be used to see the distribution, but logically, these two datasets should not be compared at all.